

ChatGPT



extract the caricullam from this:

<https://www.udemy.com/course/the-ai-engineer-course-complete-ai-engineer-bootcamp/?couponCode=MT260727G1>

Here's the extracted curriculum for [The AI Engineer Course 2026: Complete AI Engineer Bootcamp](#) .

Course Overview

- **77 sections**
- **444 lectures**
- **~29.7 hours** of video
- **107 coding exercises**
- **29 articles**
- **156 downloadable resources**
- Covers Python, NLP, Transformers, LLMs, LangChain, Hugging Face, APIs, and real-world AI engineering projects.  Udemy

Curriculum

1. Introduction to AI

- Getting started with AI
- Building your first AI application
- Structured vs. unstructured data
- Data collection
- Labeled vs. unlabeled data
- Metadata
- AI fundamentals
- Machine Learning
- Deep Learning
- Supervised learning
- Unsupervised learning
- Reinforcement learning
- AI ethics

- Future of AI
 - AI career paths
 - AI Engineer role
 - AI Developer role
 - AI Strategist role
 - AI tech stack
 - Generative AI concepts
 - Practical challenges in Generative AI
 - Robotics
 - Computer Vision
 - Hugging Face ecosystem
 - LangChain
 - AI evaluation tools  Udemy+1
-

2. Python Programming

- Why Python
- Installing Anaconda
- Setting up Jupyter Notebook
- Variables
- Data types
- Operators
- Strings
- Lists
- Tuples
- Dictionaries
- Sets
- Conditional statements
- Loops
- Functions
- Modules
- Iterators
- Error handling
- File handling
- Important Python concepts
- Best coding practices  Class Central+1

3. Natural Language Processing (NLP)

- Introduction to NLP
- Text preprocessing
- Tokenization
- Stop-word removal
- Stemming
- Lemmatization
- Part-of-Speech tagging
- Named Entity Recognition
- Sentiment Analysis
- Text vectorization
- TF-IDF
- Word embeddings
- Practical NLP applications  Class Central

4. Transformers

- Transformer architecture
- Attention mechanism
- Encoder-decoder models
- Hugging Face Transformers
- Using pretrained models
- Fine-tuning models
- Inference
- Speech-to-text applications  Udemy

5. Large Language Models (LLMs)

- What are LLMs?
- Prompt engineering
- Foundation models
- OpenAI APIs
- Model capabilities
- Practical LLM applications
- Limitations
- Safety considerations  Udemy

6. Hugging Face

- Hugging Face Hub
- Model discovery
- Pipelines
- Datasets
- Model inference
- Fine-tuning workflows
- Open-source AI ecosystem  Udemy

7. LangChain

- LangChain fundamentals
- Chains
- Prompts
- Memory
- Tools
- Agents
- Output parsers
- Building AI workflows
- Connecting LLM components  Udemy

8. APIs and AI Integration

- Working with APIs
- Connecting foundation models
- Authentication
- Making API requests
- AI application integration
- Building production-ready AI apps  Udemy

9. Vector Databases

- Embeddings
- Similarity search
- Pinecone
- Vector indexing

- Retrieval workflows  Udemy
-

10. Retrieval-Augmented Generation (RAG)

- RAG fundamentals
 - Retrieval pipelines
 - Context injection
 - Knowledge bases
 - Improving LLM responses  Udemy
-

11. Practical AI Projects

- Business case studies
 - Building AI tools
 - End-to-end AI applications
 - Interview-ready portfolio projects
 - Production workflows  Udemy
-

Skills You'll Gain

- Python programming
- Machine Learning fundamentals
- Deep Learning
- Natural Language Processing
- Transformers
- Hugging Face
- LangChain
- OpenAI APIs
- Prompt Engineering
- Vector Databases
- Retrieval-Augmented Generation (RAG)
- AI application development
- AI deployment
- AI engineering best practices  Udemy+1

Overall, the course progresses from **AI fundamentals** → **Python** → **NLP** → **Transformers** → **LLMs** → **LangChain** → **RAG** → **APIs** → **production AI applications**, making it suitable as an end-to-end AI engineering curriculum.

ChatGPT is AI and can make mistakes.